

Supplementary Material for Backbone-Modular Neuro-Symbolic Post-Correction for OCT Denoising

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This document holds the material that does not fit inside the page limit of the main paper. It adds nothing to the claims. It gives the closed form of every clinical property, the proof of the second theorem, the training algorithm, the full list of constants, the visual comparison for all four backbones, and a note on a defect found and repaired during revision.

S1. Closed form of the six clinical properties

Table S1 gives the score and the failure map of every property, and the expanded definitions of every subterm follow. All quantities are computed from the noisy input and the backbone output alone. No clean reference is used at any point.

Expanded subterms in Table ??.

P_1 . $c_{\text{cont}} = \langle \text{erode}(\text{dilate}(b_e)), b_e \rangle / (\|b_e\|_1 + \varepsilon)$, $c_{\text{efr}} = \exp(-5|r_e - 0.2|)$ with $r_e = \|\mathbf{1}[e(x) > 1.5\bar{e}]\|_1 / (\|\mathbf{1}[e(x) < 0.5\bar{e}]\|_1 + \varepsilon)$, and $c_{\text{ang}} = \exp(-2 \text{mean}(\sigma_5(\theta(x))))$.

P_2 . $s_{\text{range}} = \text{sigm}(10(\Delta_r - 0.3))$, $\Delta_r = \max(x) - \min(x)$. $s_{\text{cnr}} = \text{sigm}(15(\bar{\sigma} - 0.05))$, $\bar{\sigma} = \text{mean}(\sigma_9(x))$. $s_{\text{var}} = \max(0.2, \text{sigm}(-20(v_\sigma - 0.02) + 2))$, $v_\sigma = \text{Var}(\sigma_9(x))$. $s_{\text{cov}} = \text{sigm}(5(\gamma - 0.4))$, $\gamma = \text{mean}(\mathbf{1}[\sigma_9(x) > 0.3\bar{\sigma}])$.

P_3 . $F = \mathbf{1}[e(x) < 0.3\bar{e}]$, $q_{\text{var}} = 1 - \langle F, \sigma_5(x) / (\sigma_5(y) + \varepsilon) \rangle / (\|F\|_1 + \varepsilon)$, and $q_{\text{res}} = \exp(-5 \text{mean}|y - x|)$.

P_4 . s_{self} is the average Pearson correlation between x and shifts $(0, 2)$, $(2, 0)$, and $(2, 2)$. $s_{\text{reg}} = \exp(-12 \text{mean}(e(e(x))))$.

$s_{\text{vert}} = \text{clip}(\max|\nabla_z v(x)| / (8 \text{mean}|\nabla_z v(x)|), 0, 1)$.

P_5 . $z = \log(y + \varepsilon) - \log(x + \varepsilon)$, $cv = \sigma_{15}(|z|) / (\mu_{15}(|z|) + \varepsilon)$, and $cv^*(d)$, where d is the axial depth index, is the depth-wise Gamma-K target coefficient of variation used by the implementation.

P_6 . $s_{\text{clr}} = \text{mean}(\text{clip}(\max|\nabla_z v(x)| / (10 \text{mean}|\nabla_z v(x)|), 0, 1))$, $s_{\text{dep}} = \text{clip}(20 \text{Var}(v(x)), 0, 1)$, and $s_{\text{smo}} = 1 - \text{clip}(10 \text{mean}(\text{Var}_z(\text{mean}_w e(x))), 0, 1)$.

These are the implemented predicates. The maps $\mathbf{m}_i$ are the spatial quantities consumed by the negotiator and potential head, while the scores s_i drive the rule activations. The released code computes exactly these quantities. The relevant class is named `EnhancedGTFreePredicates` and the speckle property is computed by `PhysicsAccurateSpecklePredicate`.

S2. Proof of the contrast margin result

This is the proof of Theorem 2 of the main paper, repeated here in full. Write $\mu_{\mathcal{T}}$ and $\mu_{\mathcal{B}}$ for the mean over tissue and over background, and $\sigma_{\mathcal{B}}$ for the standard deviation over background. Let $\mathbf{b}$ be the backbone output, $\mathbf{q}$ the candidate, and $\hat{\mathbf{x}} = (1 - w)\mathbf{b} + w\mathbf{q}$ the output with $w \in (0, 1]$.

Theorem S1 (Contrast margin). Let $\eta_{\mathcal{T}} = \mu_{\mathcal{T}}(\mathbf{q}) - \mu_{\mathcal{T}}(\mathbf{b}) \geq 0$ and let $\eta_{\mathcal{B}} \geq \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b})$. If $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ then

$$\text{CNR}(\hat{\mathbf{x}}) \geq \frac{(\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta_{\mathcal{B}})}{\sigma_{\mathcal{B}}(\mathbf{b})}. \quad (\text{S1})$$

Proof:

The mean is linear, so over any fixed set of pixels the mean of a convex combination is the same convex combination of the means. Applying this over the tissue set and over the background set,

$$\begin{aligned} \mu_{\mathcal{T}}(\hat{\mathbf{x}}) &= (1 - w)\mu_{\mathcal{T}}(\mathbf{b}) + w\mu_{\mathcal{T}}(\mathbf{q}) = \mu_{\mathcal{T}}(\mathbf{b}) + w\eta_{\mathcal{T}}, \\ \mu_{\mathcal{B}}(\hat{\mathbf{x}}) &= \mu_{\mathcal{B}}(\mathbf{b}) + w\eta'_{\mathcal{B}}, \end{aligned}$$

where $\eta'_{\mathcal{B}} = \mu_{\mathcal{B}}(\mathbf{q}) - \mu_{\mathcal{B}}(\mathbf{b}) \leq \eta_{\mathcal{B}}$ by hypothesis. Subtracting,

$$\mu_{\mathcal{T}}(\hat{\mathbf{x}}) - \mu_{\mathcal{B}}(\hat{\mathbf{x}}) = (\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b})) + w(\eta_{\mathcal{T}} - \eta'_{\mathcal{B}}) \geq (\mu_{\mathcal{T}}(\mathbf{b}) - \mu_{\mathcal{B}}(\mathbf{b}))$$

which lower bounds the numerator.

For the denominator, every background pixel of $\hat{\mathbf{x}}$ is the same convex combination of the corresponding pixels of $\mathbf{b}$ and $\mathbf{q}$. Treating the background pixel values as a random variable and applying Minkowski's inequality to the centred variables,

$$\sigma_{\mathcal{B}}(\hat{\mathbf{x}}) \leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{q}) \leq (1 - w)\sigma_{\mathcal{B}}(\mathbf{b}) + w\sigma_{\mathcal{B}}(\mathbf{b}) = \sigma_{\mathcal{B}}(\mathbf{b})$$

using the hypothesis $\sigma_{\mathcal{B}}(\mathbf{q}) \leq \sigma_{\mathcal{B}}(\mathbf{b})$ in the second step. Dividing the lower bound on the numerator by the upper bound on the denominator gives the claim. Since $w > 0$ in every branch of the blending rule, the bound is at least the baseline contrast whenever $\eta_{\mathcal{T}} > \eta_{\mathcal{B}}$. ■

Section VII of the main paper reports how often each of the three hypotheses holds on the test set, so the

TABLE S1. Closed form of the six clinical properties. For each predicate, the table lists its clinical role, image-level score, and pixel-level failure map. larger $\mathbf{m}_i$ means stronger evidence that the corresponding property is locally unsatisfied. The expanded subterm definitions follow immediately below. Here $\bar{e} = \text{mean}(e(x))$, $b_e = \mathbf{1}[e(x) > \bar{e}]$, $v(z) = \text{mean}_w z$ is a depth profile, ρ_h is 1-pixel horizontal autocorrelation, and Var_k is local variance over a $k \times k$ window.

P_i	Clinical role	Aggregate score s_i	Failure map $\mathbf{m}_i$	τ_i
P_1	Edge continuity	$0.4 c_{\text{cont}} + 0.3 c_{\text{efr}} + 0.3 c_{\text{ang}}$	$(1 - \bar{e})(1 - b_e)$, $\bar{e} = e(x)/(e_{\text{max}} + \varepsilon)$	0.6
P_2	Local contrast recovery	$0.20 s_{\text{range}} + 0.35 s_{\text{cnr}} + 0.20 s_{\text{var}} + 0.25 s_{\text{cov}}$	$[\text{ReLU}(\bar{\sigma} - \sigma_9(x))/(\bar{\sigma} + \varepsilon) + 0.3 \mathbf{1}[\sigma_9(x) < 0.01]]_{[0,1]}$	0.5
P_3	Flat-region smoothness	$0.6 q_{\text{var}} + 0.4 q_{\text{res}}$	$F \odot \sigma_5(x)/(\max \sigma_5(x) + \varepsilon)$	0.75
P_4	Structural coherence	$0.4 s_{\text{self}} + 0.3 s_{\text{reg}} + 0.3 s_{\text{vert}}$	$e(e(x))/(\max e(e(x)) + \varepsilon)$	0.5
P_5	Speckle conformity	$0.7 \text{mean}(\exp(- cv - cv^* /0.15)) + 0.3 \exp(-5 \rho_h(z))$	$1 - \exp(- cv - cv^* /0.15)$	0.3
P_6	Layer visibility	$0.4 s_{\text{clr}} + 0.3 s_{\text{dep}} + 0.3 s_{\text{smo}}$	$\text{Var}_7(e(x))/(\max \text{Var}_7(e(x)) + \varepsilon)$	0.7

Algorithm 1 Training the wrapper.

Require: frozen backbone f_θ , paired training set, number of epochs E

- 1: initialise the wrapper parameters, set each property threshold to its literature value
- 2: for $e = 1$ to E do
- 3: for each batch of crops do
- 4: compute the backbone output, with no gradient to θ
- 5: compute the confidence map from the backbone features
- 6: compute the six property scores and failure maps
- 7: compute the gain, edge and strength proposals
- 8: compute the five rule truth values
- 9: compute the allocation
- 10: compose the candidate, then evaluate the objective on it
- 11: update the wrapper parameters with AdamW
- 12: end for
- 13: if e reaches the midpoint then activate the fidelity term
- 14: end if
- 15: end for

reader can see when the guarantee is active rather than assuming it always is.

S3. Training procedure and the full list of constants

A. Algorithm

Algorithm 1 states the procedure. The backbone is frozen throughout and receives no gradient. The first stage activates only the clinical terms, so the correctors learn a useful direction. The second stage activates the fidelity term, so they learn how far to go.

TABLE S2. Structural constants. None of these was tuned.

Constant	Value	Role
gate steepness	20	sharpness of the background gate
failure sharpness	10	sharpness of the score to failure conversion
smoothing window	3×3	smoothing of the brightening term
erosion window	5×5	keeps background smoothing away from tissue
crop size	96×96	training crops
batch size	4	training batch
edge bound	± 0.08	nominal bound of the additive branch
quarter resolution	$H/4 \times W/4$	the gain network runs at this scale

B. Structural constants

The main paper lists the constants that carry a design decision. Table S2 lists the remaining ones, which are structural and were not tuned.

C. What the paired data is used for

Training uses paired data. The clean reference enters the fidelity term, the clinical term and the supervision of the edge branch. Testing uses no clean image anywhere in the correction path. The six properties are computed from the noisy input and the backbone output only, which is what allows them to run on a scanner the method has never seen. The clean reference is used to score results, as in any denoising study, and never to produce them.

S4. Visual comparison for all four backbones

Fig. S1 repeats the visual comparison of the main paper for every backbone rather than the two extremes. The layout is the same. Every cell is a zoomed region chosen automatically as the window with the strongest layer structure in the clean reference, because the differences

that matter sit at layer boundaries and are invisible at full scale. The last column shows the change the wrapper applied, on a scale symmetric about zero, so the reader can see where the correction acted rather than inferring it.

S5. A defect found during revision, and its repair

We record this because the released code contains a switch that restores the earlier behaviour, and a reader who finds that switch deserves to know what it is for.

A. What was wrong

In the version of this work that was first submitted, the allocation was formed by adding the rule contributions to a base level and then clipping the result to the unit interval. The positive contributions could reach 0.884 and 0.762 on top of a base of 0.818, so the value arriving at the clip could be as large as 2.46. The clip was therefore active at every pixel of every image and the allocation was the constant one.

Two consequences follow. The rule layer had no influence on any output, since multiplying by one changes nothing. And the derivative of a clip at its bound is zero, so no gradient reached the rule parameters and they stayed at their initial values for the whole of training. We confirmed both directly, by reading the allocation map out of the trained model and by comparing every rule parameter in the saved weights against its initialiser after 2082 optimiser steps.

B. The repair

The allocation is now a bounded weighted vote. Writing $r_k \in [0, 1]$ for the truth value of rule k and $w_k \in (0, 1)$ for its learned weight,

$$a = B_{\min} + (B_{\max} - B_{\min}) \sigma(\beta) - C_1 w_1 r_1 + C_2 w_2 r_2 + C_3 w_3 r_3 + C_4 w_4 r_4 - C_5 w_5 r_5, \quad (\text{S2})$$

with $B_{\min} = 0.25$, $B_{\max} = 0.50$ and budgets $C_1 = 0.15$, $C_2 = 0.25$, $C_3 = 0.15$, $C_4 = 0.10$ and $C_5 = 0.10$. The budgets satisfy

$$B_{\max} + C_2 + C_3 + C_4 = 1, \quad B_{\min} - C_1 - C_5 = 0, \quad (\text{S3})$$

so the value lies in the unit interval before any clipping, for every input and every parameter setting. Clipping is therefore never active, saturation is impossible rather than unlikely, and the gradient reaches every rule parameter.

The repair also tightens the theory. Because the allocation is affine in the rule truth values with signed coefficients and no clipping intervenes, the Lipschitz constant of Theorem 1 of the main paper is attained rather than merely an upper bound.

C. Reproducing the earlier behaviour

Setting the environment variable
LEGACY_SATURATING_ALLOCATION to 1

restores the original computation exactly. We verified this against a copy of the pre revision source on twenty random inputs, comparing the allocation maps for equality and the rule traces for agreement. Both matched. The earlier numbers can therefore be recovered from the released code, which is why the switch exists.

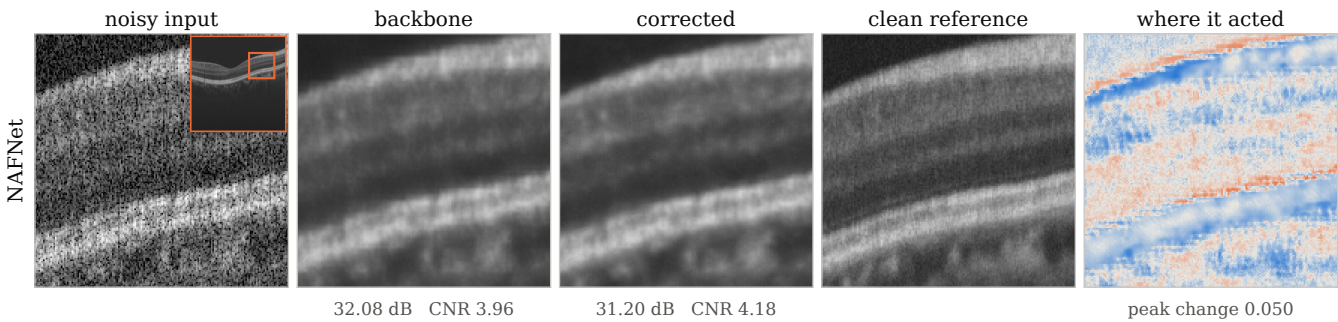


FIGURE S1. Visual comparison on PKU37 for all four backbones. Columns are the noisy input, the backbone output, the corrected output, the clean reference, and the change applied. Blue means the wrapper lowered the value and orange means it raised it.